

TOHMAJARVI KEMIE, Finland

WMO: 028320

Lat: 62.235N

Lon: 30.347E

Elev: 295

StdP: 14.54

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.0	-14.4	-27.1	1.2	-20.8	-19.7	1.9	-14.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	80.9	66.3	77.3	64.2	73.7	62.2	69.1	77.3	66.6	74.3	64.4	71.1	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.2	97.5	73.6	63.6	89.0	70.4	61.4	82.1	67.8	33.4	77.4	31.4	74.6	29.7	71.2	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-23.9	83.8	7.7	5.4	-29.4	87.7	-34.0	90.8	-38.3	93.9	-43.9	97.8		
(5)			-24.6	71.6	7.5	4.0	-30.0	74.5	-34.4	76.8	-38.6	79.1	-44.0	82.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.2	13.4	17.9	24.5	36.2	48.6	55.5	61.5	57.9	49.4	37.6	30.6	23.6
	DBStd	18.13	14.89	13.05	9.65	6.65	7.75	6.70	6.13	5.74	6.06	7.02	8.03	11.75
	HDD50	5264	1135	900	790	414	119	24	3	7	84	387	583	818
	HDD65	9845	1600	1320	1255	863	510	291	142	230	468	850	1033	1283
	CDD50	943	0	0	0	1	75	190	359	251	66	1	0	0
	CDD65	51	0	0	0	0	1	7	34	9	0	0	0	0
	CDH74	724	0	0	0	0	66	134	404	120	0	0	0	0
	CDH80	134	0	0	0	0	5	22	87	20	0	0	0	0
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	PrecAvg	26.5	1.9	1.6	1.4	1.2	1.8	2.8	2.9	3.2	2.4	2.6	2.4	2.3
(15) Precipitation	PrecMax	35.4	4.0	3.3	2.6	2.8	4.3	5.2	6.3	6.1	7.0	5.8	4.3	5.0
	PrecMin	21.1	0.5	0.2	0.2	0.2	0.7	1.3	0.5	1.0	0.9	0.6	0.4	0.9
	PrecStd	3.5	0.8	0.8	0.6	0.6	0.9	1.1	1.5	1.3	1.5	1.2	0.9	0.9
	PrecStd	3.5	0.8	0.8	0.6	0.6	0.9	1.1	1.5	1.3	1.5	1.2	0.9	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	37.3	39.2	48.2	63.8	80.5	83.0	86.8	82.9	69.1	54.4	48.4	42.4	42.4
	0.4% MCWB	36.1	36.2	38.7	49.3	59.6	64.6	70.5	68.8	60.9	51.6	46.9	41.6	41.6
	2% DB	35.4	36.3	41.9	56.7	75.2	77.9	82.2	77.2	64.7	51.8	44.4	38.7	38.7
	2% MCWB	34.5	34.3	35.8	45.9	58.0	63.9	68.3	65.4	58.0	49.6	43.2	37.7	37.7
	5% DB	33.3	34.5	38.7	52.3	69.9	73.6	78.6	73.0	61.7	49.8	42.6	36.3	36.3
	5% MCWB	32.4	33.3	34.3	42.8	56.1	60.8	66.6	63.1	56.7	47.5	41.9	35.3	35.3
	10% DB	32.1	33.0	36.2	48.5	65.0	69.8	74.8	69.5	59.3	47.8	40.4	34.6	34.6
	10% MCWB	31.5	32.2	33.2	40.8	52.4	58.8	64.3	61.5	55.8	45.9	39.6	33.8	33.8
	0.4% WB	36.5	36.6	39.6	51.4	63.2	70.3	73.2	70.9	61.9	52.8	47.1	41.9	41.9
	0.4% MCDB	36.8	38.2	44.4	58.4	74.5	77.9	83.1	79.3	66.6	54.4	48.2	42.4	42.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	34.7	34.9	36.9	47.4	59.7	65.9	70.2	66.6	59.8	50.4	43.5	37.8	37.8
	2% MCDB	35.3	36.0	40.4	56.1	70.4	74.7	79.0	73.9	63.1	51.5	44.2	38.5	38.5
	5% WB	33.0	33.5	35.2	44.0	57.0	62.5	68.1	64.8	57.9	48.1	41.8	35.6	35.6
	5% MCDB	33.6	34.3	37.7	50.7	67.8	69.3	76.2	71.2	60.5	49.0	42.4	36.1	36.1
	10% WB	31.7	32.4	33.8	41.4	54.1	60.0	65.9	62.8	56.2	46.0	39.7	34.0	34.0
	10% MCDB	32.4	33.1	35.7	47.0	62.8	68.2	73.1	68.4	58.7	47.6	40.3	34.4	34.4
(35) Mean Daily Temperature Range	MDBR	11.5	12.4	16.4	18.0	22.2	19.8	19.8	18.3	14.3	10.2	7.0	7.8	7.8
	5% DB	5.8	8.8	17.2	26.0	31.7	27.5	26.6	25.4	19.0	12.9	8.7	6.5	6.5
	5% WB	5.8	8.1	12.2	15.5	16.4	14.4	14.0	15.2	13.1	10.9	8.2	6.2	6.2
	MCDBR	5.5	7.9	12.8	22.2	26.8	23.6	23.8	21.6	15.7	10.8	8.3	6.4	6.4
	MCWBR	5.6	7.6	9.7	13.8	15.0	13.6	13.6	13.4	11.7	9.9	7.9	6.3	6.3
	MCWBR	5.6	7.6	9.7	13.8	15.0	13.6	13.6	13.4	11.7	9.9	7.9	6.3	6.3
(40) Clear-Sky Solar Irradiance	taub	0.241	0.290	0.316	0.355	0.355	0.349	0.376	0.359	0.337	0.317	0.259	0.216	0.216
	taud	2.021	2.101	2.166	2.296	2.438	2.483	2.421	2.451	2.503	2.398	2.117	1.974	1.974
	Ebn at Noon	132	205	247	263	274	278	266	261	246	202	125	80	80
	Edh at Noon	11	22	30	33	31	30	32	28	23	17	10	6	6
(44) All-Sky Solar Radiation	RadAvg	66	246	711	1167	1560	1685	1609	1229	689	267	68	31	31
	RadStd	9	59	96	126	166	103	149	138	87	38	12	4	4

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)		+0.97	N/A	N/A	N/A	N/A	N/A	-333	N/A	N/A	N/A

Nomenclature: See separate page